3.2 Existing metrics and our desiderata

We review a subset of existing metrics commonly used in the literature and discuss them in light of the above desiderata. We selected metrics that are consistent with our previously stated base assumptions. In particular, they can all be formalized within the same framework for specific functions h, g, and M, as shown in Table 1. While this list of metrics is by no means exhaustive, we think that the selected metrics are representative of the main ideas and assumptions underlying most existing metrics in the literature. Table 1 provides a condensed overview of the alignment between these metrics and our desiderata.

Average popularity lift. Abdollahpouri et al. [3] use the lift of group average popularity as a metric to quantify popularity bias. Their metric is based on groups of users. For the sake of simplicity, we focus on overall average popularity (i.e., assuming only one single group). Average popularity at the user level u (respectively, at the recommender level r) is defined as

$$AP(u) = \frac{\sum_{a \in p_u} \phi(a)}{|p_u|}, \qquad AP(r) = \frac{\sum_{a \in p_r} \phi(a)}{|p_r|}.$$

Importantly, no transformation of the raw scores is performed, that is, g is just the identity and the score aggregation function is given by the empirical mean. The final metric, average popularity lift, is then given by

$$\Delta AP(r,u) = \frac{AP(r) - AP(u)}{AP(u)}. \tag{1}$$

This metric, unfortunately, is at odds with several of our desiderata. Regarding Desideratum (1), note that the above metric is given as an empirical average of the samples of a random variable, that is, the raw popularity scores. Therefore, we would expect that the average converges to its expectation, i.e., $AP(u) \to \mathbf{E}\phi(a)$ as the number of samples $|p_u| = n \to \infty$. However, if raw popularity scores follow a power law, as we argue at the top of Section 3, this might not be the case; the metric might not converge to anything meaningful even with an infinite amount of data. The same reasoning is also valid for metrics that are derived from some form of average popularity, such as those proposed in [15] and [35], which use a popularity bias metric based on empirical moments of the raw scores.

Furthermore, the metric in (1) also does not satisfy the desideratum on anti-symmetry. Consider a user with $AP(u) = 0.5$ and a recommendation with $AP(r) = 1$ Then $\Delta AP(r,u) = 1$ but $\Delta AP(u,r) = -0.5$. In other words, (1) reports relative change (popularity bias in this case is 1, or 100%, because the recommendation popularity is 100% larger than the user base popularity). Although this could be fixed by omitting the numerator, this would only fix the required anti-symmetry, not the other desiderata. Finally, the metric in (1) also weighs changes of the popularity equally, regardless of the scale, putting it at odds with our required long tail sensitivity. On the other hand the metric is monotonic, satisfying Desideratum (5).

Gini index. The Gini index [23] is a global measure of inequality of a distribution. For the case of popularity, we define it by considering the set of items consumed by a user and the items surfaced by a recommender as

$$Gini(r) = \sum_{i=1}^{|p_r|} \left(\frac{2i - |p_r| - 1}{|p_r|} \right) \left(\frac{\phi(a_i)}{\sum_{a_i \in p_r} \phi(a_i)} \right),$$

$$Gini(u) = \sum_{i=1}^{|p_u|} \left(\frac{2i - |p_u| - 1}{|p_u|} \right) \left(\frac{\phi(a_i)}{\sum_{a_i \in p_u} \phi(a_i)} \right).$$

For the specific purpose of this metric we assume that the items $\phi(a_i)$ are sorted in increasing order (i.e., $\phi(a_i) \leq \phi(a_{i+1})$), see also [4, 11]. As these measures are defined at the level of the recommender (respectively the user) we need some way of comparing these two values. We suggest to use either some form of (relative) difference or ratio:

$$\Delta Gini(r,u) = Gini(r) - Gini(u),$$

$$\%\Delta Gini(r,u) = \frac{\Delta Gini(r,u)}{Gini(u)},$$

$$\%Gini(r,u) = \frac{Gini(r)}{Gini(u)}.$$

This comparison finally amounts to comparing the inequality of popularity in the recommender profile vs. the inequality of popularity in the user profile. Note that only the first metric $\Delta Gini(u,r)$ is anti-symmetric and zero centered, which fulfills our Desiderata (2) and (3). The metric $Gini(r)$ is well defined in case of finite variance and reasonable estimators exist in the case of infinite variance, see [20, 56], satisfying Desideratum (1). A small change in a large popularity value has smaller impact than the change of the popularity value of a small value, which fulfills our requirement of the long tail sensitivity. However, as a measure of inequality within a distribution, the Gini index is, in general, not monotonic. Imagine a case of an equal popularity of all items. If now all popularity values of the recommender are lifted by an equal increment, the Gini index remains unchanged, as the overall inequality is not varied, which is at odds with Desideratum (5).

Popularity rank correlation. For the rank correlation of a popularity bias metric we use the definition of [62] and adjust it to our notation. The popularity rank correlation measures how strong the popularity measured by the popularity scores $\phi(a)$ of a set of items is correlated with the rank decided by a recommender. We define PRU as

$$PRU(r,u) = SRC\left[\{\phi(a) \in p_u \cap p_r\}, \mathrm{rank}_r(\{a \in p_u \cap p_r\})\right],$$

where SRC denotes the Spearman-rank correlation, $p_u \cap p_r$ denotes the intersection of items that are both part of the recommender profile and the user profile and $\mathrm{rank}_r(\cdot)$ is the function that assigns to every item a a rank decide by the recommender r. Note that this definition requires a notion of ranking by the recommender which is not required by the other metrics. Additionally, this metric intersects items from the user and recommender profile, which puts it slightly at odds with our imposed framework. The metric is well-behaved as ranks can be computed without additional requirements over user and recommender profiles. However, the metric is neither zero centered, nor anti-symmetric. This metric does not satisfy the long tail sensitivity and monotonicity desiderata as changes in the value of the popularity score have no influence as long as the rank is not changed.

Desiderata	(Group) Average popularity lift [3]	Gini index [4, 11]	Popularity rank correlation [62]	Herfindahl-Diversity [4]	Log popularity difference (ours)
(1) well behaved	✗	✓	✓	✓	✓
(2) zero centered	✓	✓	✗	✓	✓
(3) anti symmetric	✗	✓	✗	✓	✓
(4) long tail sensitivity	✗	✓	✗	✗	✓
(5) monotonicity	✓	✗	✗	✗	✓
Aggregation h	mean	Gini coefficient	identity	sum of squares	mean
Transformation g	identity	normalization by total popularity	rank	normalization by total popularity	log
Comparison M	relative difference	difference	correlation	difference	difference

Table 1: Overview of common popularity bias metrics. We choose the aggregation function M (amongst Δ, $\Delta\%$ and $\%$) such that most desiderata are satisfied, indicated by ✓.

Herfindhal index. The Herfindhal-index [27] was originally developed as a measure of economic concentration and has since been applied to measure popularity dispersion, see [4]. We define it at the level of a recommender (a user respectively) as

$$H(r) = \sum_{a \in p_r} \left(\frac{\phi(a)}{\sum_{a \in p_r} \phi(a)} \right)^2, \; H(u) = \sum_{a \in p_u} \left(\frac{\phi(a)}{\sum_{a \in p_u} \phi(a)} \right)^2.$$

The metrics are again defined at the user and recommender level, therefore we need an approach for computing a measure of popularity bias based on the two. Analogous to above we identify the following form of (relative) difference or ratio:

$$\Delta H(r,u) = H(r) - H(u), \; \%\Delta H(r,u) = \frac{\Delta H(r,u)}{H(u)}, \; \%H(r,u) = \frac{H(r)}{H(u)}.$$

The metric normalizes individual popularity scores by the sum of all scores in the profile, making it well-behaved (the metric is bounded between 0 and 1). By choosing the difference of the values, we can satisfy Desiderata (2) and (3). However, large popularity values have an overall dominant effect (due to the squaring of the values), putting it at odds with Desideratum (4), Desideratum (5) (monotonicity) is also not fulfilled.

3.3 Properties of the log popularity difference metric

We suggest a new metric, denoted *log popularity difference*, that satisfies all our desiderata. It is conceptually similar to the average popularity lift metric but drops its normalization term (to ensure anti-symmetry) and introduces a log transformation of the raw popularity values (to ensure statistical robustness). The log popularity difference metric is defined as

$$\Delta M(r,u) = \frac{\sum_{a \in p_r} \log \phi(a)}{|p_r|} - \frac{\sum_{a \in p_u} \log \phi(a)}{|p_u|}. \tag{2}$$

For data with popularity scores $\phi(a)$ following a Pareto distribution, this metric satisfies all Desiderata: Desideratum (1): it is well-behaved because the log transformation of a Pareto-distributed random variable follows an exponential distribution with well-defined mean and variance; (2): it is centered around zero; (3): the metric is anti-symmetric; and (4): changes in the tail (for high popularity values) have a weaker impact than changes for small popularity values due to the log transformation. Finally, the metric satisfies Desideratum (5) because the log transformation is monotonic itself and thus preserves the monotonicity of the average popularity aggregation function. We will use this new metric to measure popularity bias in our experiments.

4 EXPERIMENTS

We aim to answer the following research questions in our experiments:

- (RQ1) How does a recommender based on off-the-shelf LLMs compare to traditional behavioral recommenders in terms of both recommendation accuracy and popularity bias ?
- (RQ2) How effective are simple prompt-based popularity bias mitigation strategies? How do these mitigation strategies trade off popularity bias against recommendation accuracy?

We conduct our experiments in the movie recommendation domain using the 'MovieLens 10M' data set [25]. The data set contains 10 million ratings on 10,000 movies from 72,000 users. We chose this dataset for the following reasons: 1) we expect general-purpose LLMs to be fairly good off-the-shelf movie recommenders, given the vast amount of movie-related content on the internet including reviews and discussions; and 2) we use the *LensKit* [17] library to evaluate standard recommendation algorithms on the task and compare popularity bias and relevance of the LLM recommender to such classical methods. We start by defining the LLM-based recommender.

4.1 WOrld Knowledge recommender (WOK): A simple LLM-based movie recommender

We use a pre-trained LLM with a prompt template that simply asks to recommend a list of ten movies based on the watch history of a given user. The prompt template also contains formatting instructions so that we can reliably parse the natural-language output returned by the LLM and resolve it to a list of movie identifiers present in the MovieLens 10M dataset. This is needed to calculate relevance and popularity bias of the recommendation. Furthermore, the LLMs are instructed to not recommend movies released after

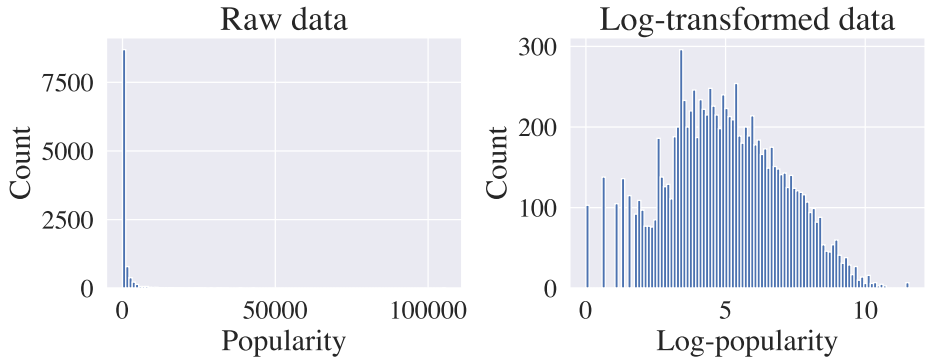

Figure 1: Popularity scores of the MovieLens dataset. Left are the raw scores, i.e., counts of how often a movie has been rated. We run a goodness-of-fit test (using the distfit package) for several heavy tailed distributions and find that a Pareto-distribution, i.e., power law, is the best fit. The estimated coefficient is $\alpha = 0.68$, makeing both mean and variance undefined. Right are log-transformed popularity scores.

2008, as these movies are not contained in the MovieLens data set, as well as movies that a user has already watched.

We call the resulting model *WOrld Knowledge recommender* (WOK) because it relies entirely on the world knowledge (also known as parametric knowledge) acquired during pre-training, i.e., it has not explicitly been trained on the MovieLens 10M dataset to recommend movies. We use various off-the-shelf LLM-APIs from Anthropic and OpenAI as back-end LLMs and call the resulting models WOK-*model-name* for different specific models during our experiments. The Appendix contains further details on API parameter choices.

4.2 Baselines

In order to study the relationship between predictive performance and popularity bias we chose a range of simple baselines to compare our WOK recommender with. *ItemKNN:* A traditional collaborative filtering algorithm, K-nearest neighbors based on item similarities. As recommended in the literature [16] we pick $K = 30$, i.e., we consider the 30 most similar items for each item. *UserKNN:* Traditional collaborative filtering based on K-nearest neighbors based on user similarities [50, 53]. As before we pick $K = 30$, i.e., we consider the 30 most similar users for each test user. *Top Popularity Recommender:* Always recommend the k most popular items. *Random Recommender:* Recommend k items uniformly at random.

4.3 Experimental Setup

We follow a standard evaluation protocol, where the ratings of each user are split into a training and testing set. The movies in the training set are used to train the algorithm, or in the case of the LLM recommender, as an input to the model at run-time (see Section 4.1). The recommender then generates a slate of recommendations. To measure the recommendation accuracy we report top five hit rate (HR@5) and top ten hit rate (HR@10).

To calculate the popularity bias of different algorithms, we approximate the raw popularity score of a movie by the total number of ratings it received. The raw as well as log-normalized scores are plotted in Fig. 1. For LLM-based recommenders, we additionally report the number of invalid recommendations. Invalid recommendations occur when the LLM violates the instructions it has received via the prompt template. This could mean that the recommended item is formatted incorrectly, has already been watched by the same user in the past, is not referring to an existing movie title, or is a movie title that does indeed exist but is not part of the MovieLens data set. Note that the baseline recommenders have access to a valid candidate set and thus will always recommend valid titles.

We repeat the experiment over five folds and report the average ± the standard error of the mean in Table 2. Each fold consists of 1000 users (i.e., we subsample the original dataset).

4.4 Results

Figure 2 and Table 2 show the results. A perfect recommender would achieve HR@5=HR@10=1 while exhibiting zero popularity bias. Regarding predictive performance, the traditional user-based collaborative filtering algorithm (UserKNN) works best. It is also among the least popularity biased models.

The lowest popularity bias, surprisingly, is achieved by the Anthropic Claude-based WOK model (WOK-claude-v2.1). In fact, only WOK-gpt-3.5 exhibits a higher popularity bias than the least-biased baseline model. Note also that the Random recommender has a strong negative popularity bias. This indicates that the watch histories of users contained in the MovieLens data set contain movies with much higher-than-average popularity scores.

We also note the low number of invalid items for the WOK models, with all models returning valid movie recommendations in at least 8.5 out of 10 cases. This means that the LLM recommenders not only correctly format the recommended movies, but also respect the query, that is, they recommend movies from valid years and not do not recommend movies that the user has already watched.

The gap in predictive performance of WOK models compared to UserKNN and ItemKNN can likely be explained by the additional information that those models have access to. While WOK models in this current implementation only access an individual test user's watch history, the collaborative filtering baselines have access to the full history of all user-item interactions. One could make